

README

Helix OTA — Master Diagrams Index

Field	Value
Revision	1
Created	2026-06-07
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Status	active
Status summary	Index of the six canonical Mermaid (.mmd) diagram sources for the Helix OTA master specs: system architecture, device update sequence, 1.0.0-MVP ER model, 1.0.1 staged rollout, deployment topology, and the build-to-device trust chain. All six were validated with mmdc (mermaid-cli) and render to SVG/PNG. Names and structure are kept consistent with the committed 1.0.0-MVP specs (same table and service names).
Issues	HelixConstitution clause numbers (§11.4.61, §7.1, §11.4.6) are carried from the corpus convention and are UNVERIFIED. staged_rollout.mmd describes the 1.0.1 engine, not the 1.0.0-MVP all-at-once path (so noted in the diagram and below). Reverse-proxy image (caddy:2), dashboard image, and image digests in deployment_topology.mmd are placeholders marked UNVERIFIED per deployment/overview.md §8.
Fixed	N/A (initial revision). Note: during validation two parser-breaking characters were corrected in update_sequence.mmd (a ?/; inside a sequence message and a ; inside a Note) so all six files now parse cleanly under mmdc.
Continuation	When scripts/export_docs.sh is extended to render standalone .mmd

Field	Value
	files (it currently extracts and renders only fenced mermaid blocks from Markdown), wire these six into the export run and commit the generated SVG/PNG under <code>_exports/</code> . Add a CI step that fails the build if any <code>.mmd</code> here fails <code>mmdc</code> validation.

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The table-of-contents requirement is mandated by HelixConstitution §11.4.61 (UNVERIFIED clause number). This document carries its ToC immediately after the metadata table, per [../documentation_standards.md §3](#).

1. Purpose and scope

This directory holds the **canonical Mermaid diagram sources** for the Helix OTA master specifications. Mermaid is the source of truth for these diagrams per [../documentation_standards.md §6](#); SVG/PNG are derived, never hand-edited.

Each `.mmd` carries a `%%` comment header naming its source-of-truth spec section, so the diagram and its prose stay in lock-step. All six are kept consistent with the committed 1.0.0-MVP specs — the **same table names** (12 MVP tables) and the **same service names** (`ota-server`, `postgres`, `minio`, `reverse proxy`, `dashboard`).

2. Diagram catalogue

File	Type	What it shows	Primary source spec
system_architecture.mmd	graph TB	Three planes (Dashboard / Control / Device) + data-plane infra + NEW <code>ota-*</code> modules, with the two deliberately extractable seams (OS-adapter, rollout-engine).	2026-06-07-helix-design.md §4
update_sequence.mmd	sequenceDiagram	Device poll → 204 (idle) / 200 (update)	

File	Type	What it shows	Primary source spec
		assigned) branch → download → device re-verify → applyPayload → reboot → boot verify → telemetry, including the boot-failure auto-rollback path.	../1.0.0-mvp/endpoints.md §12; §5
er_model.mmd	erDiagram	The 12 MVP tables — users, api_keys, devices, device_groups, device_group_members, artifacts, artifact_versions, releases, deployments, device_deployments, telemetry_events, audit_logs — with attributes and FK relationships (delete behavior annotated).	../1.0.0-mvp/database/schema.r §5
staged_rollout.mmd	flowchart TD	1.0.1 phased rollout (5 % → 10 % → 30 % → 100 %) with health-gated halt-on-failure (halt wins over advance) and operator abort/rollback. Note: 1.0.0-MVP is all-at-once; this is the 1.0.1 engine.	../1.0.1-staged-rollout/README.m master §8
deployment_topology.mmd	flowchart TB	Containerized service set (ota-server, postgres, minio, reverse proxy, dashboard) across edge / app / data networks, on the vasic-digital/containers substrate, with compose vs k8s split.	../1.0.0-mvp/deployment/overv §3–§6
trust_chain.mmd	flowchart LR	Build-pipeline sign → server verify (GATE 1) → device re-verify (GATE 2) → update_engine (GATE 3) → AVB/dm-verity (GATE 4) → A/B boot_control with native rollback. TUF/Uptane MVP-forward seam shown as deferred.	../1.0.0-mvp/security/signing_verifica §2, §4–§7; master §6

3. Rendering and export

These are **standalone .mmd source files**. They render to SVG/PNG with mermaid-cli:

```
mmdc -i system_architecture.mmd -o system_architecture.svg
mmdc -i system_architecture.mmd -o system_architecture.png
```

The corpus export pipeline `scripts/export_docs.sh` converts the Markdown corpus to HTML/DOCX (+ PDF when a LaTeX engine is present) and renders Mermaid diagrams to SVG + PNG. **Accuracy note (§7.1, no bluff):** as of this revision `export_docs.sh` extracts and renders fenced ````mermaid` blocks *embedded in Markdown*; it does not yet iterate standalone .mmd files in this directory. Those are rendered directly with mmdc (above). Wiring the standalone .mmd set into the export run is tracked in this document's Continuation row. The pipeline degrades gracefully: a missing renderer is logged and SKIPPED, never fatal (`./export_pipeline.md`).

4. Validation status

VERIFIED (executed): all six .mmd files were parsed and rendered with mmdc (/opt/homebrew/bin/mmdc, mermaid-cli) on 2026-06-07; each produced a non-empty SVG with no parse error. This is positive evidence per §7.1 — the claim “syntactically valid Mermaid” is backed by an actual successful render, not asserted.

To re-validate locally:

```
for f in *.mmd; do mmdc -i "$f" -o "/tmp/${f%.mmd}.svg" && echo
    "OK $f"; done
```

5. Consistency with the committed specs

- **Table names** in `er_model.mmd` are exactly the 12 tables defined in `../1.0.0-mvp/database/schema.md §4–§5` and created by `migrations/001_initial_schema.up.sql` — no renamed, added, or invented tables. Deferred entities (`deployment_phases`, `rollouts`, `rollback_history`, TUF metadata) are intentionally **not** shown, matching `schema.md §8`.
- **Service names** in `deployment_topology.mmd` (`ota-server`, `postgres`, `minio`, `reverse proxy`, `dashboard`) match `../1.0.0-mvp/deployment/overview.md §3`.
- **Endpoints / status codes** in `update_sequence.mmd` (`GET /api/v1/client/update` → 204/200; `POST /client/telemetry`) match the API spec §12.
- **Gate numbering** in `trust_chain.mmd` (GATE 1–4) matches `../1.0.0-mvp/security/signing_verification.md §2`.

6. Anti-bluff / UNVERIFIED register

Per §7.1 / §11.4.6 (`../documentation_standards.md §8`):

- **VERIFIED (executed):** the six .mmd files parse and render under mmdc (§4).

- **UNVERIFIED:** HelixConstitution clause numbers (§11.4.61, §7.1, §11.4.6) — carried from corpus convention, not cross-checked against the constitution text.
- **UNVERIFIED:** reverse-proxy image (caddy:2), dashboard image, and image digests in `deployment_topology.mmd` (placeholders per `deployment/overview.md` §8).
- **UNVERIFIED:** that `export_docs.sh` renders standalone `.mmd` files — it currently does not (§3); standalone render is via `mmdc` directly.
- No fabricated table names, service names, endpoints, or submodule names are introduced; gated/deferred items are marked, not invented.

7. Sources

- [`../2026-06-07-helix-ota-design.md`](#) — master design (§4 planes/seams, §5 MVP path, §6 trust model, §8 staged rollout).
- [`../documentation\_standards.md`](#) — metadata/ToC (§2–§3), diagram conventions (§6), anti-bluff (§8).
- [`../export\_pipeline.md`](#) — the real, tested export pipeline (`scripts/export_docs.sh`).
- [`../1.0.0-mvp/database/schema.md`](#) — 12 MVP tables and relationships.
- [`../1.0.0-mvp/deployment/overview.md`](#) — service set and topology.
- [`../1.0.0-mvp/api/endpoints.md`](#) — update-check (204/200) and telemetry endpoints.
- [`../1.0.0-mvp/security/signing\_verification.md`](#) — the four trust gates.
- [`../1.0.1-staged-rollout/README.md`](#) — staged-rollout engine and halt-on-failure.